Provide a list of all the GNN architectures, bechmarking datasets, and all the evaluation metrics since the beginning of the GNN's to latest research on GNN's so that I can run the models, evaluate on standard benchmarking datasaets and look for a novel comparision table and point out the scope for future directions

Standard **GNN Benchmarks** and **Future Gaps**

The attached literature-review papers frame the field as moving from classical spectral and spatial GNNs toward attention-based, transformer-integrated, spatiotemporal, multimodal, and application-specific designs, while recurring bottlenecks remain **scalability, heterogeneity, interpretability, and temporal dynamics** (Mienye & Viriri, 2025; Deng et al., 2026). Across the broader benchmark literature, the main practical message is that fair comparison still depends as much on **splits, hyperparameter tuning, and task-specific metrics** as on architecture choice (Dwivedi et al., 2020; Shchur et al., 2018; Errica et al., 2019; Hu et al., 2020).

Architecture Families

The earliest GNN lineage starts with **recurrent GNNs**, and later surveys organize the field into recurrent GNNs, convolutional GNNs, graph autoencoders, and spatiotemporal GNNs (Wu et al., 2020; Xu et al., 2018). A later survey expands that inventory to recurrent, convolutional, attention-based, autoencoding, and spatiotemporal designs in medical imaging (Mienye & Viriri, 2025).

A practical architecture list supported across the papers is below.

Family	Representative models explicitly named in the papers	Typical use
Recurrent	Early GNN, gated/recurrent variants	Node and graph representation
Spectral convolution	**GCN**, Chebyshev/spectral methods, MoNet	Node classification, graph learning
Spatial/message passing	**GraphSAGE**, MPNN-style aggregation	Inductive node learning
Attention-based	**GAT**, attention diffusion, causal-attention variants	Heterogeneous neighbor weighting
Expressive/WL-style	**GIN**	Graph classification
Autoencoding	Graph autoencoders	Representation and link prediction
Spatiotemporal	STGNN, temporal graph models	Dynamic graphs and time series

Family	Representative models explicitly named in the papers	Typical use
Transformer-integrated	Graph-transformer hybrids	Long-range and multimodal structure
Structure-aware / causal	Causal GNN variants, adaptive graph models	Robustness and generalization

FIGURE 1 Major GNN architecture families and their common use cases

The core named baselines that recur in fair evaluations are **GCN, GraphSAGE, GAT, MoNet, and GIN** (Shchur et al., 2018; Joghatayi, 2026; Job et al., 2025; Deng et al., 2026). Expressiveness work argues that **GIN** is a theoretically motivated neighborhood-aggregation model with Weisfeiler-Lehman-level power, while standard GCN and GraphSAGE can fail to distinguish some graph structures (Xu et al., 2018).

Benchmark Datasets

The benchmark ecosystem is now split between **small canonical citation/social datasets**, **graph-classification collections**, and **large realistic benchmarks**. TUDataset provides more than 120 graph-classification and regression datasets with standardized loaders and evaluation tools, while OGB provides large-scale datasets across social, biological, molecular, code, and knowledge-graph domains with unified protocols (Morris et al., 2020; Hu et al., 2020).

Common node-classification benchmarks repeatedly named in unified studies are **Cora, Citeseer, PubMed, Amazon Computers, Amazon Photo, Coauthor CS, and Coauthor Physics** (Shchur et al., 2018; Joghatayi, 2026). Link-prediction benchmarking has grown more rigorous because datasets and metrics were previously inconsistent and negatives were often unrealistically easy (Li et al., 2023).

For scaling studies, **IGB** was introduced specifically to address gaps in labeling, heterogeneity, feature flexibility, and graph size, with both homogeneous and heterogeneous academic graphs and far more labeled data than prior public large-scale datasets (Khatua et al., 2023).

Evaluation Metrics

Metrics vary by task rather than by architecture. Node and graph classification studies most often use **accuracy**, and recent benchmarking papers also emphasize **precision, recall, F-score**, plus efficiency measures such as **training time** and **memory footprint** (Shchur et al., 2018; Joghatayi, 2026; Job et al., 2025).

Graph and link tasks often use **AUC**, while some graph-regression settings use **RMSE** (Li et al., 2023; Khemani et al., 2024). OGB's main contribution is not one universal metric, but **application-specific metrics and splits** matched to each dataset (Hu et al., 2020).

Evaluation of **explainability** is now its own subfield. GRAPHXAI provides benchmark datasets, synthetic generators, and explanation metrics because older explanation datasets were often unreliable or trivially solved (Agarwal et al., 2022).

Evidence Coverage Across GNN Benchmarks

	Node Tasks	Graph Tasks	Link Tasks	Large Scale	Fairness
Architectures	GAP	GAP	GAP	GAP	GAP
Datasets	GAP	GAP	GAP	GAP	GAP
Metrics	GAP	GAP	GAP	GAP	GAP
Explainability	GAP	GAP	GAP	GAP	GAP
Scalability	GAP	GAP	GAP	GAP	GAP

The clearest gap is not a lack of architectures, but a lack of **standardized cross-task comparison**. The benchmark papers repeatedly show that ranking changes under different splits, tuning procedures, and dataset choices, and that no single model dominates across datasets (Dwivedi et al., 2020; Shchur et al., 2018; Errica et al., 2019). That is why apparently stronger models often lose to simpler ones when evaluation is controlled carefully (Shchur et al., 2018; Errica et al., 2019; Fung et al., 2021).

A second major gap is that **large-scale, heterogeneous, temporal, and explanation-aware** benchmarks are still underdeveloped relative to citation-network and medium-scale graph tasks (Khatua et al., 2023; Hu et al., 2020; Yuan et al., 2020; Deng et al., 2026). A strong novel comparison table would therefore compare architectures across **task type, inductive setting, scalability cost, robustness to splits, explainability support, and OOD realism**, rather than only reporting raw accuracy.

In short, the best usable starting set is **GCN, GraphSAGE, GAT, GIN, MoNet, and one spatiotemporal or transformer-hybrid baseline**, run on **Cora/Citeseer/PubMed, Amazon, Coauthor, TUDataset, OGB, and IGB**, with task-matched metrics and fixed reproducible splits. The strongest future direction is a benchmark that jointly measures **accuracy, efficiency, robustness, scale, and explanation quality** across the same evaluation pipeline.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at https://consensus.app. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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